2023

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level A
Middle Primary
Grades **3–4**

DATE

23 September

TIME ALLOWED

60 minutes

Instructions and Information 考试说明

1. You must maintain silence at all times.
   考试过程中请保持安静。
2. Do not open the paper until told to do so.
   收到监考老师指示前，请勿打开试卷。
3. Mobile phones are not allowed. If you have one you must give it to a supervisor now.
   禁止使用手机。如果携带手机，请立刻交给监考老师。
4. You must not leave your seat during the competition.
   考试期间禁止离开座位。
5. You may not borrow equipment without a supervisor's permission.
   未经监考老师许可，禁止向他人借用工具。
6. If you wish to leave the room a supervisor must accompany you.
   如果想离开考场，必须有监考老师陪同。
7. If you have any other questions, raise your hand and wait for a supervisor.
   如果有其他问题，请举手示意监考老师。
8. You are allowed a working time of 60 minutes. There is no extra reading time
   本场考试时长为 60 分钟，没有额外的读题时间。
9. There are no penalties for incorrect answers. You should attempt all questions.
   答错不扣分，请尝试回答所有问题。
10. The questions have been thoroughly checked. Each question stands as written and no further explanation can be provided.
    本场考试的所有问题已仔细核查，不提供任何口头解释。
11. Diagrams are NOT drawn to scale. They are intended only as aids.
    图形未按比例绘制，仅供参考。
12. You may use calculators and printed language dictionaries.
    允许使用计算器和纸质语言词典。

Important: No student may sit the competition more than once or sit more than one division of the competition. The AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

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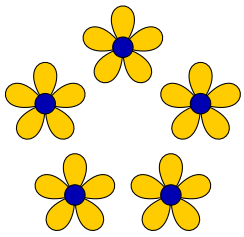
Middle Primary Division

Questions 1 to 10 are worth 3 marks each.

1-10 题，每题 3 分

1. What is the total number of petals on all 5 flowers?

请问图中 5 朵花一共有多少片花瓣？



(A) 10 (B) 15 (C) 20 (D) 25 (E) 50

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2. $2 + 3 + 7 + 8 =$

(A) 10 (B) 20 (C) 30 (D) 40 (E) 50

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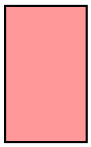
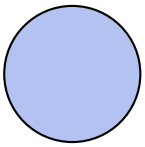
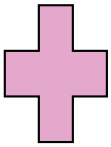
3. Which one of these shapes is a rectangle?

以下哪个是长方形？

(A)  (B) (C) (D)  (E) 

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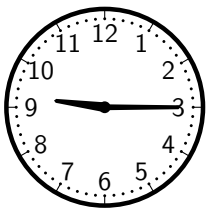
4. Which digital clock time matches the time shown on the clock face?

以下哪个数字时钟显示的时间与钟面上的时间相同？



(A) 3:09 (B) 9:03 (C) 9:15 (D) 10:03 (E) 10:15

---

5. Emma has a bag containing 5 red, 4 yellow, 1 black and 2 blue buttons. When she chooses 1 button at random, what colour is it most likely to be?

小艾有一袋纽扣，里面装有 5 个红色纽扣，4 个黄色纽扣，1 个黑色纽扣和 2 个蓝色纽扣。她从中随机选出一个纽扣，这个纽扣最有可能是什么颜色？

(A) green 绿色 (B) blue 蓝色 (C) black 黑色 (D) yellow 黄色 (E) red 红色

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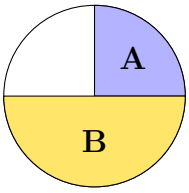
6. What fraction of the circle is part A?

请问 A 部分占整个圆的几分之几？



(A) one-half 二分之一

(B) one-third 三分之一

(C) two-thirds 三分之二

(D) one-quarter 四分之一

(E) three-quarters 四分之三

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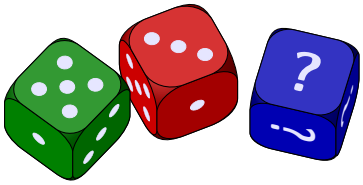
7. In a board game, Nik rolls three standard dice, one at a time. He needs his three rolls to add to 12. His first two dice rolls are 5 and 3. What does he need his third roll to be?



一个棋盘游戏中，小尼投掷三枚标准的骰子，每次掷出一枚。他需要三次掷出的点数之和为 12。

他前两次掷出的点数分别是 5 和 3，请问小尼第三次需要掷出的点数是多少？

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

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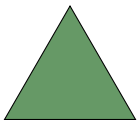
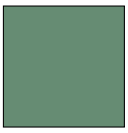
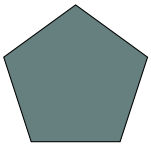
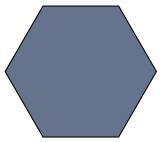
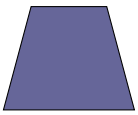
8. Which shape has *only one* line of symmetry?

以下哪个图形只有一条对称轴？

(A)  (B)  (C)  (D)  (E) 

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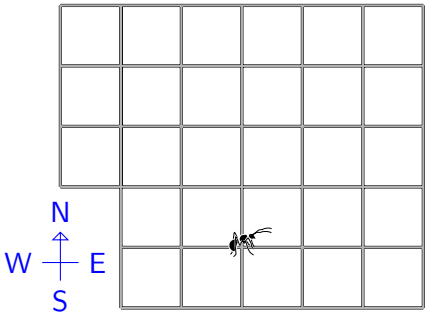
9. Anastasia Ant is on a sheet of wire mesh with 10 cm squares. She can only move along the wires. She moves 10 cm east, then 30 cm north and 20 cm west.

What is the least possible distance she needs to move to return to her starting point?

如图所示，蚂蚁小安在一张由 10 cm 小方格构成的钢丝网上。她只能沿着钢丝网移动。她先向东移动 10 cm，然后向北移动 30 cm，最后向西移动 20 cm。

请问如果小安要回到起点，需要移动的最小距离是多少？



(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm (E) 50 cm

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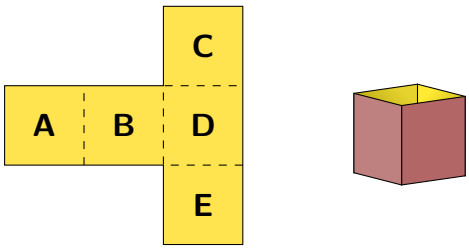
10. A piece of card is cut out and labelled as shown in the diagram.

It is folded along the dotted lines to make a box without a top.

Which letter is on the bottom of the box?

将一张卡片裁剪成如图所示的形状，并标记上字母，然后将其沿着虚线折叠成一个没有顶部的盒子。

请问哪个字母位于盒子底部？



(A) A (B) B (C) C (D) D (E) E

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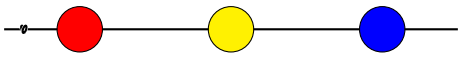
Questions 11 to 20 are worth 4 marks each.
11-20 题，每题 4 分

11. Jiang is threading beads onto a string to make a necklace. He decides to use a sequence, starting with a red bead, then a yellow bead and a blue bead. He repeats this pattern until he has 20 beads on the necklace.

How many red beads will he use?

小江正在将珠子串在一根线上来制作项链。他决定从一颗红色珠子开始，然后依次是一颗黄色珠子和一颗蓝色珠子。他不断重复这个顺序，直到项链上有 20 颗珠子。请问小江会用到多少颗红色珠子？



(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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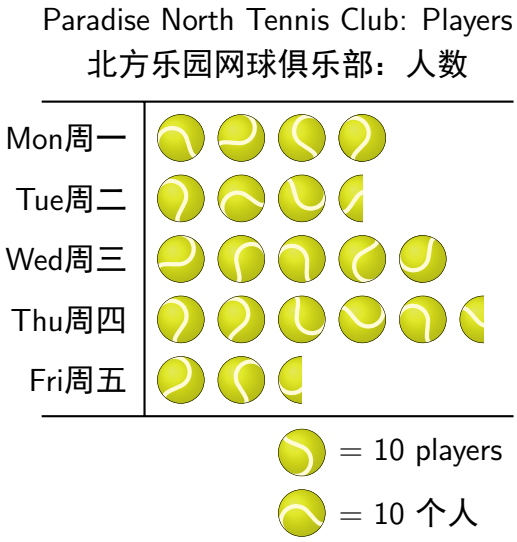
12. How many people played tennis on the busiest day of the week at the Paradise North Tennis Club?

请问北方乐园网球俱乐部一周中最繁忙的一天有多少人打网球？

(A) 45 (B) 50 (C) 55 (D) 60 (E) 65



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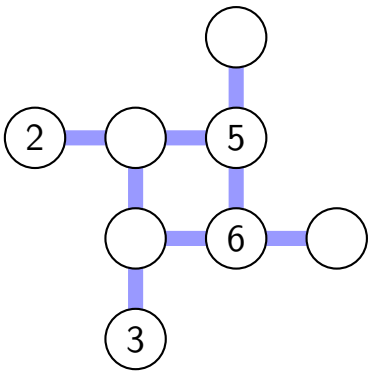
13. When this puzzle is complete, each of the four lines will add up to 14. What will the four missing numbers add to?

右侧的谜题完成后，四条直线中每条直线上的数字相加之和为 14。请问四个缺失的数字相加之和为多少？

(A) 10 (B) 14 (C) 15 (D) 16 (E) 18



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14. Which one of the following could give an answer of 25?

请问以下哪个选项的答案可以是 25？

(A) an odd number plus an odd number
一个奇数加上一个奇数

(B) an even number multiplied by an even number
一个偶数乘以一个偶数

(C) an even number divided by an odd number
一个偶数除以一个奇数

(D) an odd number minus an even number
一个奇数减去一个偶数

(E) an even number multiplied by an odd number
一个偶数乘以一个奇数

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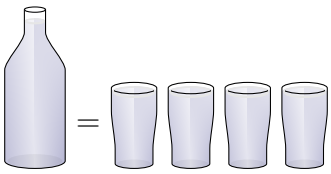
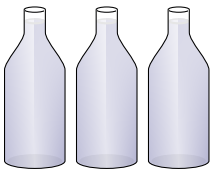
15. This bottle holds 4 glasses of water.

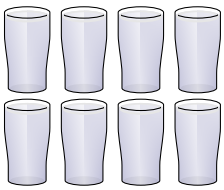
一个瓶子可以装 4 杯水。

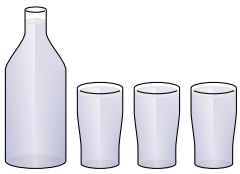


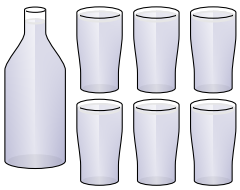
Which one of the following holds the most water?

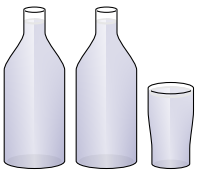
以下哪项装的水最多？

(A)


(B)


(C)


(D)


(E)


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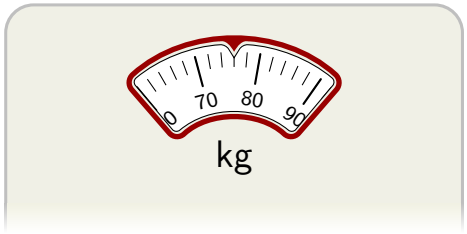
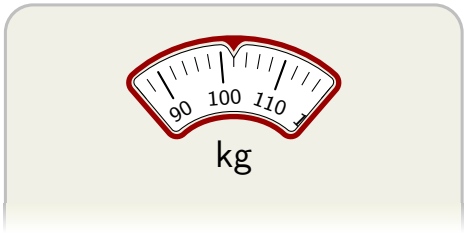
16. Fido needs to be weighed at the vet, but he won't sit still. So the vet weighs himself and then weighs himself holding Fido.

小费需要在兽医诊所称重，但它坐不稳。因此，兽医先称了自己的重量，然后抱着小费再次称重。





How much does Fido weigh?

请问小费有多重？

(A) 23 kg (B) 26 kg (C) 28 kg (D) 30 kg (E) 32 kg

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17. Grandma adds 3 spoons of sugar to the juice of 10 lemons to make lemonade. How many lemons are needed if 15 spoons of sugar are used to make Grandma's recipe?

奶奶制作柠檬饮料，会在 10 个柠檬榨的汁中加入 3 勺糖。按照奶奶的配方，如果使用 15 勺糖来制作柠檬饮料，请问需要多少个柠檬？

(A) 20 (B) 22 (C) 25 (D) 30 (E) 50

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18. Janus is making patterns using square tiles.

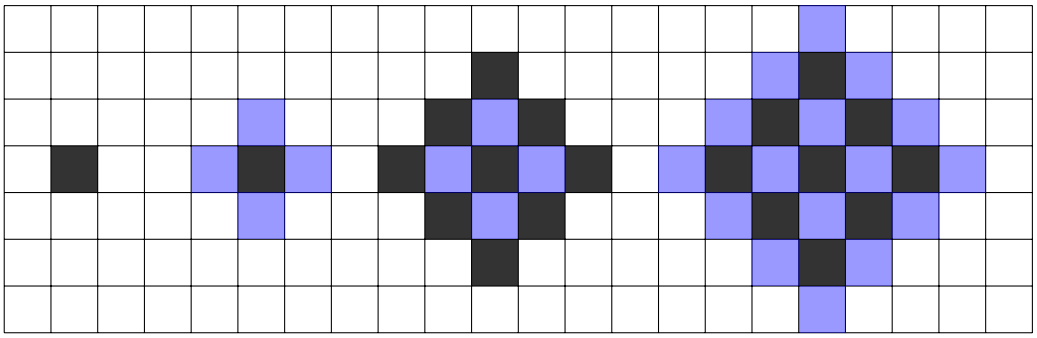
Each pattern is made by copying the previous pattern, then adding new tiles around the outside edges.

The new tiles alternate in colours, as shown.

小简正在用正方形瓷砖制作图案。

制作每个图案都是先复制上一个图案，然后沿着该图案的每条边缘各添加一块新瓷砖。

如图所示，新瓷砖的颜色交替排列。



Pattern 1 图案 1 Pattern 2 图案 2 Pattern 3 图案 3 Pattern 4 图案 4

What will be the total number of tiles in Pattern 5?

请问第 5 个图案一共有多少块瓷砖？

(A) 25 (B) 33 (C) 37 (D) 40 (E) 41

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19. My laptop's battery meter tells me what fraction of the battery charge is left. I used the laptop for 4 hours and 20 minutes and it went from

to

What would the reading be after using the laptop for another 130 minutes?

我的笔记本电池条会显示剩余电量的比例。我使用笔记本电脑 4 小时 20 分钟后，电量从

变为

如果我再使用笔记本电脑 130 分钟，请问电池条会显示为什么？

(A)

(B)

(C)

(D)

(E)

---

20. Syed's mother had some money to share with her family.

She gave one-quarter of her money to Syed.

Then she gave one-third of what was left to Ahmed.

Then she gave one-half of what was left to Raiyan.

She was left with \$15, which she kept for herself.

How much money did Syed's mother have to start with?

小赛的妈妈要分一些钱给家人。

她把四分之一的钱给了小赛，

把剩下的钱的三分之一给了小艾，

然后她再把剩下的钱的一半给了小莱，

最后还剩下 \$15。

请问小赛的妈妈最开始有多少美元？

(A) \$30 (B) \$45 (C) \$60 (D) \$90 (E) \$120

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Questions 21 to 25 are worth 5 marks each.

21-25 题，每题 5 分

21. In 2023, Janine's birthday is on a Thursday. Ngoc's birthday is one month later, meaning that it is on the same day-number in the next month. Ngoc's birthday is not on a Saturday or a Sunday. Which day of the week is Ngoc's birthday?

在 2023 年，小珍的生日是在星期四。小可的生日比小珍晚一个月，即在下个月的同一天。小可的生日不在星期六或星期日。请问小可的生日是星期几？

(A) Monday 星期一 (B) Tuesday 星期二 (C) Wednesday 星期三 (D) Thursday 星期四 (E) Friday 星期五

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22. Sally empties her piggy bank to see how much she has saved. She has 20 coins altogether and the total value of her coins is \$3.80. She notices that she has twice as many 50c coins as 20c coins. The remaining coins are 10c coins. How many 10c coins does she have?

小莉将存钱罐全部倒出来想看看存了多少钱。她一共有 20 枚硬币，总价值是 3.80 美元。50 美分硬币的数量是 20 美分硬币的两倍，其余的都是 10 美分硬币。请问她一共有多少枚 10 美分硬币？

(A) 2 (B) 4 (C) 8 (D) 11 (E) 14

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23. Three different numbers from 1 to 20 are chosen. They add to 20. Two of the numbers are 2 apart. How many possibilities are there for the largest number?

从 1 到 20 中选出三个不同的数字，这三个数字之和为 20。其中两个数字相差 2。请问最大的数字会有多少种可能？

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

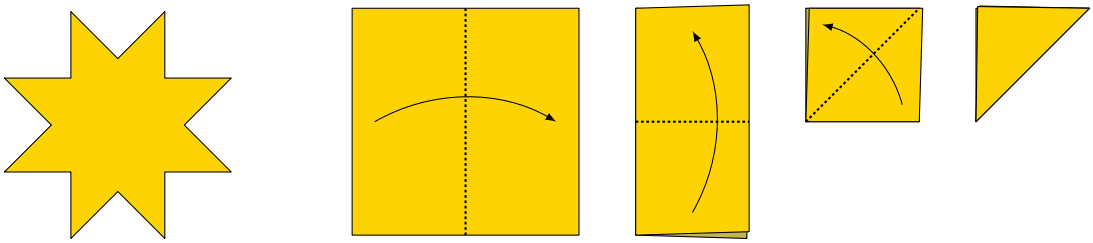
---

24. Estelle is making decorations shaped like the 8-pointed star shown.

She folds a square of paper to make a triangle with 8 layers as shown.

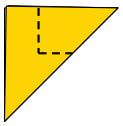
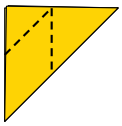
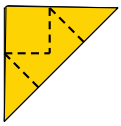
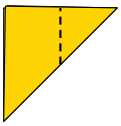
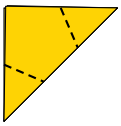
小爱正在制作八角星形状的装饰品。

如图所示，她将一张正方形纸张，折叠成有 8 层的三角形。



How could she cut the triangle so that the unfolded shape is the star?

请问小爱如何裁剪这个三角形，才能使得展开后的形状是八角星形？

(A)  (B)  (C)  (D)  (E) 

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25. Tom is digging holes in his garden for his new plants. As he is working he slows down. Each new hole takes a minute longer to dig than the last hole.

The first 5 holes have taken him 35 minutes in total. How long will it take him to dig the next 2 holes?

汤姆正在花园里为他的新植物挖坑。他工作时放慢了速度，每挖一个新坑都比上一个多花一分钟。

他挖前 5 个坑一共花了 35 分钟。请问他挖接下来的 2 个坑需要多长时间？

(A) 11 minutes 11 分钟 (B) 17 minutes 17 分钟 (C) 21 minutes 21 分钟

(D) 24 minutes 24 分钟 (E) 25 minutes 25 分钟

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For questions 26 to 30, colour in the bubbles on the answer sheet to record whole-number answers from 0 to 999. Questions 26 to 30 are worth 6, 7, 8, 9 and 10 marks respectively.

26 题至 30 题的答案为 0 至 999 的整数，请正确填涂在答题卡上。第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，第 29 题占 9 分，第 30 题占 10 分。

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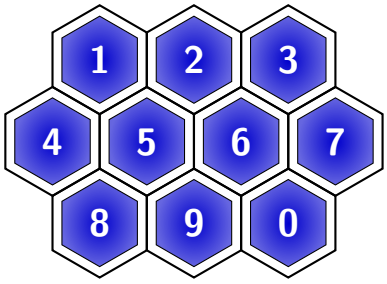
26. To unlock a security door, Daniyal enters a code on the keypad shown. To make it easy to remember:



- the code has 10 digits
- the code uses every digit exactly once
- the code steps from button to button only if the two buttons touch
- the code is as large as possible.

What are the last 3 digits in Daniyal's code?

为了打开一扇安全门，Daniyal 在如图所示的键盘上输入一串密码。为了方便记忆：

- 这个密码由 10 个数字组成
- 这个密码使用每个数字恰好一次
- 这个密码只能从一个按键上的数字到另一个相邻按键上的数字
- 这个密码的数值尽可能大

请问这个密码的最后 3 个数字是多少？

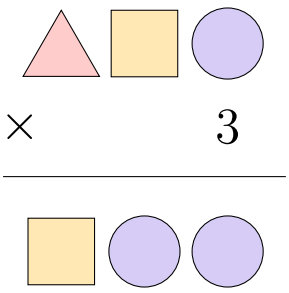
---

27. In this puzzle, △, □ and ○ represent different nonzero digits.

What is the three-digit number △□○?

在右侧谜题中，△、□ 和 ○ 表示不同的非零数字。

请问这个三位数 △□○ 是多少？



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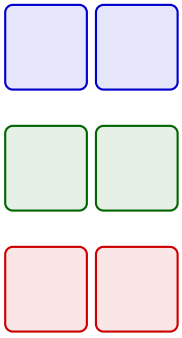
28. Ollie arranges the numbers 1 to 6 to form three 2-digit numbers.



Exactly one of his numbers divides by 3 and exactly one other divides by 4. None of them divide by 5.

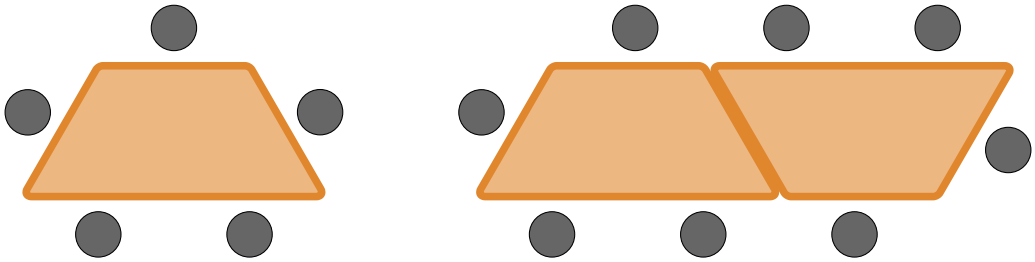
The three numbers add to 93.

What is the number which does not divide by either 3 or 4?

小奥将数字 1 到 6 进行排列，形成三个两位数。
其中恰好只有一个数能被 3 整除，恰好只有另外一个数能被 4 整除，没有能被 5 整除的数。这三个数相加之和为 93。
请问不能被 3 或 4 整除的数是多少？

---

29. A table in the shape of a trapezium can seat 5 people. When two tables are put together in a row, 8 people can be seated. What is the smallest number of trapezium tables required to seat 2023 people if they are all placed in a row?
一张梯形桌子可以坐 5 个人。当两张梯形桌子并排放在一起时，可以坐 8 个人。如果要坐 2023 人，至少需要多少张梯形桌子并排放在一起？



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30. Yifan has a construction set consisting of red, blue and yellow rods. All rods of the same colour are the same length, but differently coloured rods are different lengths. She wants to make quadrilaterals using these rods.

小伊有一个含有红色、蓝色和黄色小棒的组装套件。颜色相同的小棒长度相同，颜色不同的小棒长度也不同。她想用这些小棒拼成四边形。

- When she uses two red, one blue and one yellow rod, the perimeter of the quadrilateral is 36 cm.
- When she uses two blue, one red and one yellow rod, the perimeter is 35 cm.
- When she uses two yellow, one blue and one red rod, the perimeter is 33 cm.

- 使用两根红色、一根蓝色和一根黄色小棒拼成的四边形周长是 36 cm。
- 使用两根蓝色、一根红色和一根黄色小棒拼成的四边形周长为 35 cm。
- 使用两根黄色、一根蓝色和一根红色小棒拼成的四边形周长为 33 cm。

What number do you get when you multiply the lengths of one red rod, one blue rod and one yellow rod?

将一根红色、一根蓝色和一根黄色小棒的长度相乘，请问得到的数是多少？

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